

2082

6. Write short notes on any **three** of the following :
5×3=15

- (a) Implantation
- (b) Types & functions of glial cells.
- (c) Role of Sertoli cells
- (d) Types of proteins in skeletal muscle

[This question paper contains 4 printed pages]

Your Roll No.

: 08.06.2026(M)

Sl. No. of Q. Paper

: 2082 I

Unique Paper Code

: 2232011203

Name of the Paper

: DSC-Human Physiology
: Control and Coordination
Systems

Name of the Course

: B.Sc.(Hons.) Zoology,
(NEP-UGCF)

Semester

II

Time : 2 Hours

Maximum Marks : 60

Instructions for Candidates :

- (a) Write your Roll No. on the top immediately on receipt of this question paper.
- (b) Attempt any **four** questions in **all**. Question **NO.1** is compulsory.
- (c) Draw well labelled diagram wherever necessary.

1. (a) Define the following : (any **four**) 1×4=4

- (i) Saltatory conduction
- (ii) Latent period

- (iii) Synapse
 (iv) Sarcomere
 (v) Temporal summation
- (b) Differentiate between the following: any **three** : $2 \times 3 = 6$
- (i) Absolute vs Relative refractory period
 (ii) Graded potential and action potential
 (iii) Follicular phase vs Luteal phase
 (iv) Permissive effect and Synergistic effect
 (v) Fused and unfused tetanus
- (c) Expand the following abbreviations: $0.5 \times 4 = 2$
- (i) EPSP
 (ii) DHT
 (iii) GABA
 (iv) ICSH
- (d) Give the location and function of the following: $1 \times 3 = 3$
- (i) Nebulin
 (ii) Parafollicular Cells.
 (iii) Ependymal Cells.

2. (a) Explain the signal transduction pathway of water soluble hormones. 8
 (b) How does pancreas regulate the normal levels of glucose in the blood? 7
3. (a) Explain the muscle contraction cycle and the molecular events involved with it. 8
 (b) Describe the events occurring at the myoneural junction. 7
4. Explain menstrual cycle with respect to the uterine and ovarian cycle. Emphasize the role of different hormones regulating the cycle. 15
5. How are thyroid hormones synthesized, stored and secreted. Give an account of their physiological effects. 15
6. (a) Explain the mode of transmission of an impulse in a myelinated neuron. 9
 (b) Explain different types of glial cells and discuss their role. 6